

Started on	Saturday, 20 June 2026, 11:03 AM
State	Finished
Completed on	Saturday, 20 June 2026, 12:00 PM
Time taken	56 mins 15 secs
Grade	13.00 out of 15.00 (86.67%)

Question 1

Complete

Mark 4.00 out of 5.00

Problem

A computer vision dataset contains **20,000 images**.

The model is trained using:

- Batch Size = 250 images
- Number of Epochs = 40

Required

1. Calculate the number of batches per epoch.
2. Calculate the number of iterations per epoch.
3. Calculate the total number of iterations during the entire training process.
4. If the learning rate is reduced by a factor of 10 after epoch 20 and the initial learning rate is 0.01, what will be the new learning rate?
5. Explain why a pretrained MobileNet model may be preferred over training a CNN from scratch for this dataset.

1. The number of batches per epoch = Dataset size / Batch Size = $20000/250 = 80$ **batch per epoch**
2. Number of iterations per epoch = Number of batches per epoch = **80 iteration**
3. Total number of iterations = Number of iterations per epoch * Number of Epochs = $80*40 = 3200$ **iteration**
4. $\eta_{\text{new}} = \eta_{\text{old}}/10 = 0.01/10 = 0.001$
5. The Advantage of using pretrained model is that it's trained on huge dataset like in this case, MobileNet which trained on ImageNet (1.2 milion image, 1000 class) and it used high computation power which is hard to own such resources in personal use.

Second, The accuracy ... to get to the accuracy that the pretrained model got, we need months of tries.

So, We can conclude that using transfer learning with pretrained model will save time and cost ... just we freeze some layers and train the head in our classes and some upper layers fine tune them like in this case 20k images

Comment:

Question 2

Correct

Mark 0.50 out of 0.50

A model reaches 98% training accuracy but only 70% validation accuracy.

Which technique is MOST suitable to reduce this problem?

- ☒ a. C) Dropout ✓
- ☐ b. A) Increase epochs only
- ☐ c. D) Remove validation data
- ☐ d. B) Increase learning rate only

Your answer is correct.

The correct answer is:

C) Dropout

Question 3

Correct

Mark 0.50 out of 0.50

During Transfer Learning, a model trained on ImageNet is reused for a small dataset.

What is the recommended strategy?

- ☒ a. B) Freeze early layers and train the top layers ✓
- ☐ b. A) Train all layers from scratch
- ☐ c. D) Remove pretrained weights
- ☐ d. C) Delete convolution layers

Your answer is correct.

The correct answer is:

B) Freeze early layers and train the top layers

Question 4

Correct

Mark 0.50 out of 0.50

A network repeatedly overshoots the minimum because updates are too large.

Which hyperparameter should be adjusted first?

- ☐ a. D) Kernel Size
- ☐ b. A) Batch Size
- ☐ c. C) Number of Classes
- ☒ d. B) Learning Rate ✓

Your answer is correct.

The correct answer is:

B) Learning Rate

Question 5

Correct

Mark 0.50 out of 0.50

A researcher observes that the optimizer stops updating weights because the gradient becomes zero at a point that is neither a local minimum nor a maximum.

This point is called:

- ☐ a. A) Global Minimum
- ☐ b. B) Contour Point
- ☒ c. C) Saddle Point ✓
- ☐ d. D) Batch Point

Your answer is correct.

The correct answer is:

C) Saddle Point

Question 6

Incorrect

Mark 0.00 out of 0.50

In Depthwise Convolution, if the input has 32 channels, the number of depthwise filters should be:

- ☐ a. 3
- ☐ b. 16
- ☒ c. 1 ✖
- ☐ d. 32

Your answer is incorrect.

The correct answer is:

32

Question 7

Correct

Mark 0.50 out of 0.50

A model suffers from Internal Covariate Shift.

Which layer directly addresses this issue?

- ☒ a. C) BatchNormalization ✔
- ☐ b. A) Dropout
- ☐ c. D) Pooling
- ☐ d. B) Softmax

Your answer is correct.

The correct answer is:

C) BatchNormalization

Question 8

Correct

Mark 0.50 out of 0.50

A standard convolution is replaced with Depthwise Separable Convolution.

What is the main benefit?

- ☐ a. B) More memory usage
- ☒ b. C) Reduced computation ✓
- ☐ c. A) More parameters
- ☐ d. D) Larger feature maps

Your answer is correct.

The correct answer is:

C) Reduced computation

Question 9

Correct

Mark 0.50 out of 0.50

Nesterov Momentum differs from standard Momentum because it:

- ☒ a. B) Computes gradients at a look-ahead position ✓
- ☐ b. D) Uses only squared gradients
- ☐ c. A) Uses no momentum term
- ☐ d. C) Removes learning rate

Your answer is correct.

The correct answer is:

B) Computes gradients at a look-ahead position

Question 10

Correct

Mark 0.50 out of 0.50

You are training a CNN and the validation loss has stopped improving for 10 epochs.

Which callback is specifically designed to reduce the learning rate automatically?

- ☒ a. C) ReduceLROnPlateau ✓
- ☐ b. A) CSVLogger
- ☐ c. B) ModelCheckpoint
- ☐ d. D) BatchNormalization

Your answer is correct.

The correct answer is:

C) ReduceLROnPlateau

Question 11

Correct

Mark 0.50 out of 0.50

A model is trained using batch size = 1.

Which optimization strategy is being used?

- ☐ a. D) Momentum Gradient Descent
- ☒ b. C) Stochastic Gradient Descent ✓
- ☐ c. A) Batch Gradient Descent
- ☐ d. B) Mini-Batch Gradient Descent

Your answer is correct.

The correct answer is:

C) Stochastic Gradient Descent

Question 12

Correct

Mark 0.50 out of 0.50

A CNN is trained with:

- Batch Normalization
- Dropout = 0.5
- Adam Optimizer
- Data Augmentation
- Transfer Learning (ImageNet)

Training and validation accuracies both remain around **52%** after 50 epochs.

What is the MOST likely explanation?

- ☐ a. **B)** Vanishing gradients in lower layers.
- ☐ b. **D)** Internal Covariate Shift.
- ☐ c. **A)** Overfitting due to insufficient regularization.
- ☒ d. **C)** Underfitting caused by excessive regularization and/or too many frozen layers. ✓

Your answer is correct.

The correct answer is:

C) Underfitting caused by excessive regularization and/or too many frozen layers.

Question 13

Correct

Mark 0.50 out of 0.50

A deep neural network is being trained for medical image classification. Training loss decreases very slowly and gradients in early layers are almost zero. What is the most likely problem?

- ☐ a. A) Overfitting
- ☐ b. D) Underfitting
- ☒ c. B) Vanishing Gradient ✓
- ☐ d. C) Data Leakage

Your answer is correct.

The correct answer is:

B) Vanishing Gradient

Question 14

Correct

Mark 0.50 out of 0.50

Which statement about transfer learning is TRUE?

- ☐ a. B) It starts training from random weights.
- ☐ b. D) It only works with CNNs.
- ☒ c. C) It transfers knowledge from a related task ✓
- ☐ d. A) It requires identical output classes.

Your answer is correct.

The correct answer is:

C) It transfers knowledge from a related task

Question 15

Correct

Mark 0.50 out of 0.50

A dataset contains only 500 images. You generate rotated, flipped, and shifted versions.

The primary purpose is:

- ☒ a. C) Improve generalization ✓
- ☐ b. B) Reduce image size
- ☐ c. D) Increase epochs
- ☐ d. A) Increase GPU memory

Your answer is correct.

The correct answer is:

C) Improve generalization

Question 16

Correct

Mark 0.50 out of 0.50

Which callback should be used if you want to save only the best-performing model during training?

- ☐ a. A) CSVLogger
- ☐ b. D) EarlyStopping
- ☒ c. B) ModelCheckpoint ✓
- ☐ d. C) Lambda Layer

Your answer is correct.

The correct answer is:

B) ModelCheckpoint

Question 17

Correct

Mark 0.50 out of 0.50

A CNN detects edges in the first convolution layer.

Which filter type is responsible?

- ☒ a. C) Edge Detector ✓
- ☐ b. B) Shape Detector
- ☐ c. A) Color Detector
- ☐ d. D) Blob Detector

Your answer is correct.

The correct answer is:

C) Edge Detector

Question 18

Correct

Mark 0.50 out of 0.50

Which optimizer combines the ideas of Momentum and RMSProp?

- ☒ a. C) Adam ✓
- ☐ b. A) AdaGrad
- ☐ c. D) Nesterov
- ☐ d. B) SGD

Your answer is correct.

The correct answer is:

C) Adam

Question 19

Correct

Mark 0.50 out of 0.50

A team wants a lightweight CNN for mobile devices with minimum computations.

Which architecture is MOST suitable?

- ☐ a. B) ResNet50
- ☐ b. A) VGG19
- ☒ c. C) MobileNet ✓

Your answer is correct.

The correct answer is:

C) MobileNet

Question 20

Correct

Mark 0.50 out of 0.50

A researcher is training a CNN for bone tumor classification using the following observations:

- Training Accuracy = **99%**
- Validation Accuracy = **71%**
- Validation Loss starts increasing after Epoch 15.
- Gradient values are stable (neither vanishing nor exploding).
- The dataset contains only **1,200 images**.
- The model is trained from scratch using **ResNet50** with all layers trainable.

Which combination of actions is MOST likely to improve validation performance?

- ☐ a. **D)** Increase model depth by adding more convolutional layers and remove Batch Normalization.
- ☐ b. **A)** Increase learning rate and increase epochs to 300.
- ☒ c. **B)** Apply Data Augmentation, add Dropout, and use Transfer Learning with frozen lower layers. ✓
- ☐ d. **C)** Replace ReLU with Sigmoid in all layers and reduce batch size to 1.

Your answer is correct.

The correct answer is:

B) Apply Data Augmentation, add Dropout, and use Transfer Learning with frozen lower layers.

Question 21

Incorrect

Mark 0.00 out of 0.50

A CNN receives an image of a dog. The dog's position changes slightly, but the model still classifies it correctly.

- ☐ a. A) Fully Connected Layer
- ☐ b. C) Softmax Layer
- ☐ c. B) Pooling Layer
- ☒ d. D) Flatten Layer ✗

Your answer is incorrect.

The correct answer is:

B) Pooling Layer